Ferran Barba Sáez

+34 627 265 633 // ferranbarbasaez@gmail.com // LinkedIn // GitHub // Portfolio

SUMMARY

I am a hardworking and motivated programmer, focused on analysing, learning and creating. My experience working has given me both the ability to be independent and to carry myself in a team environment. I have a proactive attitude towards my work, and I boast a high adaptability and a will to learn new things. I really enjoy programming and I am looking to leverage my previous knowledge and experience as a developer.

EXPERIENCE

UI/UX and Gameplay Programmer, New Rites Studio (Valencia)

September 2025 - June 2026

Nohmen: Blood Rites (Fixed Cameras, Action, UE5, C++, BP)

- Design and code of the entirety of the UI using the CommonUI Plugin, including the development of core screens such as Main Menu, Player HUD, and all the settings screens.
- Implemented minor adjustments and fixes in gameplay features to ensure seamless UI integration.

Nohmen: Blood Rites - Trailer

Fullstack Developer, Ceteck Tecnológica S.L. (Valencia)

June 2021 - August 2023

Zero Defects Manufacturing Platform (HTML, CSS, C#, .NET, Java, JavaScript, Angular)

- Led the end-to-end development of two ZDMP project applications focused on industrial machine optimization.
- Delivered high-impact frontend and backend features, successfully refactoring the existing web architecture to maximize performance, scalability, and code maintainability.

ZDMP - Zero Defects Manufacturing Platform

Backend Developer Internship, Journify (Valencia)

March 2020 - May 2023

Journify (HTML, CSS, C#, Java, JavaScript, Angular)

- Backend Engineer for a workforce transit optimization mobile app.
- Led the architectural refactoring of the platform from a monolith into a decoupled microservices structure, isolating endpoints by domain functionality.

PROJECTS

MentatEngine (C++, Lua, OpenGL, ImGui, Android)

September 2025 - June 2026

- Game engine made from scratch using C++ and OpenGL backend, with a compatibility layer to add Vulkan in the future.
- The main features are Entity Component Sytem, Frustrum Culling with Octrees, Mesh and Texture Streaming, four Rendering Pipelines, Multithread Job System and a port to Android.

MentatEngine Demo

Ferran Barba Sáez

+34 627 265 633 // ferranbarbasaez@gmail.com // LinkedIn // GitHub // Portfolio

Jimbo Delta Tower (Puzzles, Platformer, C++, UE5, BP)

January 2025 - June 2025

- 3D puzzle-platformer where the protagonist Jimbo tries to climb the tallest building in the city, dodging guards to test his newly gifted hang glider.
- Developed the push and drag mechanic, map design and build, UI/UX and visual aspect

Jimbo Delta Tower - Game Trailer

Railgun Defender: Aegis (Tower Defense, C#, Unity, Mobile)

January 2025 - June 2025

- Rogue like tower defense on space. Defend your core and upgrade your railgun to defeat several types of enemies and bosses.
- Contributed to enemies design and behaviour, procedural background, power up mechanics and UI/UX

Railgun Defender: AEGIS - Game Trailer

EDUCATION

HND in Computing: Video Games Programming

September 2023 - June 2026

ESAT Valencia (Escuela Superior de Arte y Tecnología) **- Overall: Distinction**

- Specialized in mid and low-level software architecture and high-performance game development.
- Core focus on C++ and memory management, 3D mathematics, physics implementation, and graphics programming.
- Experienced in working within multidisciplinary teams (programmers, artists, and designers) following professional production pipelines and agile methodologies.

Senior Technician in Multiplatform Application Development

September 2018 - June 2020

Progresa Formación Profesional - Valencia

- Focused on multi-platform software development.
- Specialised in JavaScript and Java for backend and Android/Web environments.

SKILLS

- C & C++
- C#
- LUA
- ARM
- SQL
- Unreal Engine 5
- Unity 3D
- OpenGL

LANGUAGES

- Spanish
- English
- Catalan